

THIS WINDOW INSERT KEEPS WINTER HEAT INSIDE

A clear panel in a light wood frame with a strip of foam, pressed into the trim. About 30 dollars a window, no holes in the house, and a renter can do it.

On a cold night, the biggest hole in a cold room is the one you look through. A single pane window, three feet by five, pours out about 175 watts of heat all night, like three old 60 watt bulbs glowing out into the dark. The insert puts a sealed pocket of still air in front of the glass and cuts that loss roughly in half. You never touch the window itself.

THE HONEST NUMBERS

175 watts

Bare single glass, 15 square feet, 68 inside and 28 outside. A 40 degree spread on an ordinary winter night.

88 watts

The same window with a sealed layer of still air in front of it. Cut in half. A new double pane with the coating gets to about 53.

12 dollars

Saved per window per winter on gas heat, at about 2,500 cold hours, 7.5 therms, a 90 percent furnace, and a dollar forty five a therm.

40 dollars

Saved per window per winter on electric baseboards. Ten leaky windows on baseboards is 400 dollars a season. A heat pump lands around a third of that.

25 to 50%

How much a sealed extra layer over the glass cuts window heat loss. Windows carry about a quarter of a home's heating and cooling energy.



Foam squashes against the jamb, the panel stays put by friction, and the handle pulls it straight back out.

WHY A SHEET OF FILM WORKS

The film only has one job: holding air still. Still air is a very good insulator, and every puffy jacket and fiberglass batt is mostly a way of trapping it. Make the gap thinner than about half an inch and heat conducts straight across. Make it much wider and the air rolls: warm air rises on the room side, cools on the glass side, sinks, and the loop carries heat across. The sweet spot for a vertical gap is around half an inch to five eighths. Your frame holds three quarters of an inch of still air, and a double pane window is the same trick in glass. The deep second gap between panel and glass still counts, because the foam sealed it shut. A sealed pocket, even a deep one, beats a leaky window every time.

MEASURE, FRAME, SKIN, SEAL

Four things: a clear skin, a frame, foam tape, and a tape measure. Staples go in your frame, never in the house.

10 to 20 dollars

Shrink film kit. The crinkly five window kit. One kit gives you about two double sided panels. Twin wall polycarbonate is the tough upgrade, same frame.

about 3 dollars a stick

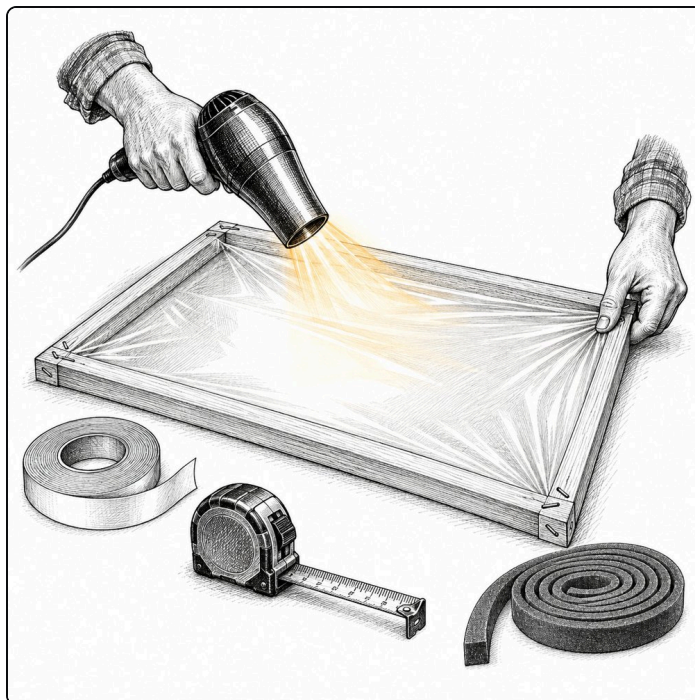
One by two furring strips. The cheapest pine at the lumber yard. Laid flat, the frame ends up three quarters of an inch deep.

quarter inch

Foam weatherstrip. Quarter inch thick, three quarters wide, the grey closed cell kind that will not soak up water. The frame size only works with quarter inch foam, so buy quarter inch.

CUT LIST AND BUILD STEPS

- 1 Look at the window.** The panel sits inside the wood jamb, in front of the sash. You need about an inch of flat jamb, clear of the sash lock and blind brackets.
- 2 Measure width** at the top, middle, and bottom, and **height** on both sides. Old openings are not square.
- 3 Use the smallest of each, then take a quarter inch off both.** That is your frame size.
- 4 Cut the two long pieces** to the full frame height. **Cut the two short pieces** 3 inches shorter than the frame width, so they fit between the long ones.
- 5 Glue each corner** and drive two staples through it, or one small screw with a pilot hole first.
- 6 Measure both diagonals.** When they match, the frame is square. Over 4 feet tall, add one strip across the middle so it cannot bow.
- 7 Stretch film across one face,** stick it down with the kit's double sided tape, trim the edge. Flip and do the other face.
- 8 Warm it gently with a hair dryer,** keep the dryer moving, until both skins pull tight as a drum.
- 9 Run the foam around the outside edge** of the frame.
- 10 Screw a small cabinet handle** to the wood on one side and tug it hard. That is your grip.
- 11 Wipe the sill dry** and check it for rot or black spots before you cover it.
- 12 Press the panel in.** The foam seals the whole perimeter. Friction holds it. The handle pulls it out.



Frame flat, film on both faces, hair dryer to tighten, foam around the edge.

WHY THE COLD CHAIR IS COLD

Cold does not flow anywhere. Heat does. Your body gives off about 100 watts and about half of it radiates toward anything colder than your skin. At night the glass sits close to 40 degrees, the coldest surface in the house, so you sit in that chair and shine at it all evening. The thermostat says 68 and it is not lying. The insert puts a surface in the fifties between you and the glass, a wall you can see through, and that side of you stops bleeding heat at the same thermostat setting.

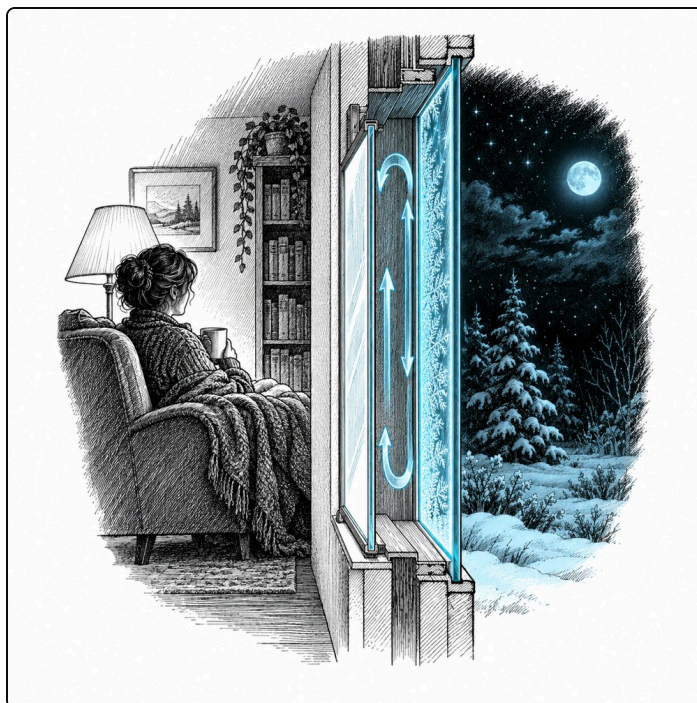
Run your hand around the edge on a windy night. Every draft that used to whistle in around the sash now stops at your foam.

MOLD, EGRESS, HEAT, PAINT

Match the rung to the room. Do it on a Saturday before the first hard frost, not the week the bill arrives.

LINES YOU DO NOT CROSS

- ☐ **Egress.** Never screw or glue a panel over a bedroom window. Every bedroom needs a window a person can open and climb out of, no tools, no keys, in the dark, in a fire. The friction fit pulls out in seconds by the handle. Pull it out once with your kids in daylight. If anyone sleeping in that room cannot pull it easily, leave that window clear. Make sure the sash behind it still opens.
- ☐ **Mold.** Warm room air drops its water as fog on cold glass, and mold can start in about two days. Keep the seal tight on the warm side, tighter than the window's own weatherstripping, so any damp dries toward the cold side. Run 30 to 40 percent humidity when it is cold out. Install only on a dry frame and sill. Check behind the panel every day the first week and after every cold snap. Anything wet: pull it, dry everything, check the foam, the film edges, and the room humidity.
- ☐ **Heat.** Shrink film is not flame resistant. A radiator or electric baseboard under the window has a clearance to anything burnable. If the panel would sit inside it, leave that window out. Never park a space heater or a candle near either one.
- ☐ **Combustion air.** A gas water heater, furnace, or wood stove with an open flue needs air to breathe. Seal every window in a leaky old house at once and you can starve that flue and pull exhaust back into the room. Have a technician test the draft before and after you tighten up. Carbon monoxide alarms on every level and outside each bedroom.
- ☐ **Paint.** House built before 1978? Do not sand or scrape the sashes to make a fit. That old paint can be lead. Do not force the panel against painted trim. Peeling jamb: get the paint checked first.



The sealed gap holds still air. Even where it rolls, a sealed pocket beats a leaky window.

THE LADDER, PER 3 BY 5 WINDOW

3 to 6 dollars

Bare film taped to the trim. One winter, ugly, and it works.

18 to 30 dollars

The framed double film above. Reusable for years. Pays for itself in one winter on electric, two or three on gas.

55 to 85 dollars

Same frame, twin wall polycarbonate. Buy the 3 by 6 sheet, ribs vertical, screw it on, solid tape on the top ends, vented tape on the bottom. Rigid, many winters.

about 510 dollars

Commercial acrylic insert with a compression tube, the kind Indow makes. Cuts street noise roughly in half.

about 500 dollars

Replacement double pane, installed, 250 to over a thousand each. Lasts 30 years. Ten windows is 5,000 dollars to save about 550 a year on baseboards or 170 on gas: 9 winters to break even on electric, about 30 on gas.

Match the rung to the room. Bedroom and any window that gets bumped: polycarbonate. Spare room and basement: framed film. The living room chair: the commercial insert, if you have it. Curtains help at night but most do not seal; air rolls down behind them and the loop carries on.

Watch the full video: youtube.com/@SketchySurvival101

More free sheets: sketchysurvival101.com/free

Subscribe so the next cold snap does not catch you flat footed.